

# No proper operator ideal has the Berger–Coburn phenomenon on Hardy or Bergman space

Candidate full-solution packet for arXiv:2304.09716, Open Problem 1

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**Status: candidate full solution, likely valid, subject to human review.** Under the standard meaning of a proper two-sided operator ideal, the answer to Open Problem 1 is no. Every proper ideal between infinite-dimensional Hilbert spaces consists of compact operators. The source paper’s analytic-symbol examples therefore rule out the Berger–Coburn phenomenon for every proper ideal at once.

## 1 The open problem

Let  $X$  be the Hardy space  $H^2(\mathbb{T})$  or the Bergman space  $A^2(\mathbb{B}_n)$ , let  $Y$  be the corresponding  $L^2$  space, and let

$$H_f = (I - P)M_f : X \longrightarrow Y$$

be the Hankel operator with bounded symbol  $f$ . The source says that an operator ideal  $\mathcal{M} \subset \mathcal{B}(X, Y)$  has the Berger–Coburn phenomenon (BCP) when

$$H_f \in \mathcal{M} \iff H_{\bar{f}} \in \mathcal{M} \quad (f \in L^\infty).$$

After exhibiting failure for compact operators, Hu and Virtanen ask [1, Open Problem 1]:

Are there nontrivial operator ideals in  $\mathcal{B}(H^2(\mathbb{T}), L^2(\mathbb{T}))$  or in  $\mathcal{B}(A^2(\mathbb{B}_n), L^2(\mathbb{B}_n))$  for which the Berger–Coburn phenomenon holds?

Here “nontrivial” is read in the standard sense “proper,” excluding the full class  $\mathcal{B}(X, Y)$ , for which membership is vacuous. The answer is no.

Figure 1 reproduces printed page 4, including both the decisive analytic-symbol examples and Open Problem 1.

One reason which makes Berger and Coburn's compactness result so striking is that analogous statements in the settings of the Hardy space and the Bergman space fail as seen in the following examples.

**Example 2.** Let  $b$  be the Blaschke product

$$b(z) = \prod_{k=1}^{\infty} \frac{\alpha_k - z}{1 - \bar{\alpha}_k z} \frac{|\alpha_k|}{\alpha_k} \quad (z \in \mathbb{D}),$$

where  $\alpha_k = 1 - \frac{1}{2^k}$ . Because  $b$  is a bounded analytic function,  $H_b = 0$  is trivially compact, but it can be shown that  $\bar{b}$  is not in the little Bloch space and hence  $H_{\bar{b}}$  is not compact according to Axler's characterization of compact Hankel operators with conjugate analytic symbols (see pp. 318–319 of [3] and the related references therein). Thus, there is no BCP for  $K(A^2, L^2(\mathbb{D}))$ . Another example in  $A^2$  to the same effect can be obtained from the bounded analytic function  $f(z) = \exp\left(\frac{z+1}{z-1}\right)$  defined for  $z \in \mathbb{D}$ .

The counterexample  $b$  above can be modified to show that there is no BCP for  $K(A^2, L^2(\mathbb{B}_n))$ . In fact, we can simply extend  $b$  by setting  $b(z) = b(z_1)$  for  $z = (z_1, z_2, \dots, z_n) \in \mathbb{B}_n$ . Then  $b$  is a bounded holomorphic function on  $\mathbb{B}_n$  and  $H_b = 0$ . On the other hand,

$$\lim_{|z| \rightarrow 1} (1 - |z|) \frac{\partial b(z)}{\partial z_1} \neq 0$$

so that  $b$  is not in the little Bloch space as Theorem 16 in [16] shows. Now Theorem E of [4] implies that  $H_{\bar{b}} \notin K(A^2, L^2(\mathbb{B}_n))$ .

To see that there is no BCP for  $K(H^2, L^2(\mathbb{T}))$ , we can choose a function  $f$  in  $H^\infty \setminus VMO$  (using a result of Wolff [12] on the class  $QA = VMO \cap H^\infty$ ), so that  $H_f = 0$  is compact while  $H_{\bar{f}}$  cannot be compact because of the characterization that states  $H_f$  and  $H_{\bar{f}}$  are both compact if and only if  $f \in VMO$ .

It may be reasonable to ask the following question.

**Open Problem 1.** Are there nontrivial operator ideals in

$$B(H^2(\mathbb{T}), L^2(\mathbb{T})) \quad \text{or in} \quad B(A^2(\mathbb{B}_n), L^2(\mathbb{B}_n))$$

for which the Berger-Coburn phenomenon holds?

We now turn our attention back to Fock spaces. While we already know this from the result of Berger and Coburn, it is still worth pointing out that examples in the spirit of Example 2 are doomed to fail because if  $f$  is a bounded entire function, then it is constant and hence  $H_{\bar{f}}$  is also compact. In fact, this is an indication of why there is BCP

Figure 1: The source counterexamples and Open Problem 1.

## 2 Idea of the proof

There are only two ingredients. First, a proper two-sided operator ideal between infinite-dimensional Hilbert spaces cannot contain a noncompact operator. Indeed, every noncompact  $T$  is bounded below on some infinite-dimensional closed subspace. After identifying the domain with that subspace,  $T$  has a bounded left inverse, so the identity factors through  $T$ . The ideal property then puts every bounded operator in the ideal.

Second, the source paper already gives a bounded analytic symbol  $b$  in each Hardy/Bergman setting such that  $H_b = 0$  while  $H_{\bar{b}}$  is noncompact. The zero operator belongs to every ideal, whereas the lemma excludes  $H_{\bar{b}}$  from every proper ideal. Thus the BCP equivalence fails for all proper ideals simultaneously.

## 3 The Hilbert-space ideal lemma

For clarity, by an operator ideal in  $\mathcal{B}(X, Y)$  we mean a linear subspace  $\mathcal{M}$  satisfying

$$ATB \in \mathcal{M} \quad (T \in \mathcal{M}, A \in \mathcal{B}(Y), B \in \mathcal{B}(X)).$$

This is exactly the rectangular component of a standard two-sided operator ideal.

**Lemma 1.** *Let  $X, Y$  be separable infinite-dimensional Hilbert spaces. If  $\mathcal{M} \subsetneq \mathcal{B}(X, Y)$  is an operator ideal, then*

$$\mathcal{M} \subseteq \mathcal{K}(X, Y).$$

*No norm-closure hypothesis is required.*

*Proof.* Suppose  $T \in \mathcal{M}$  is not compact. Write  $|T| = (T^*T)^{1/2}$ . There is an  $\varepsilon > 0$  for which the spectral subspace

$$E = \mathbf{1}_{[\varepsilon, \infty)}(|T|)X$$

is infinite-dimensional. Otherwise all these spectral subspaces would be finite-dimensional, and the finite-rank operators  $|T|\mathbf{1}_{[1/m, \infty)}(|T|)$  would converge in norm to  $|T|$ , making  $T$  compact.

Choose an isometry  $V : X \rightarrow X$  with range  $E$ . Then

$$\|TVx\| = \||T|Vx\| \geq \varepsilon\|x\|,$$

so  $TV$  has closed range and a bounded left inverse. More explicitly, define  $S : Y \rightarrow X$  by  $S(TVx) = x$  on  $\text{ran}(TV)$  and by zero on its orthogonal complement. Then  $STV = I_X$ .

For arbitrary  $R \in \mathcal{B}(X, Y)$ ,

$$R = (RS)TV.$$

Since  $T \in \mathcal{M}$  and  $RS \in \mathcal{B}(Y)$ ,  $V \in \mathcal{B}(X)$ , the ideal property gives  $R \in \mathcal{M}$ . Hence  $\mathcal{M} = \mathcal{B}(X, Y)$ , contradicting properness. Every member of a proper ideal is therefore compact.  $\square$

## 4 Negative answer to Open Problem 1

**Theorem 1.** *Let*

$$(X, Y) = (H^2(\mathbb{T}), L^2(\mathbb{T})) \quad \text{or} \quad (X, Y) = (A^2(\mathbb{B}_n), L^2(\mathbb{B}_n)).$$

*No proper operator ideal  $\mathcal{M} \subsetneq \mathcal{B}(X, Y)$  has the Berger–Coburn phenomenon for all bounded symbols. Thus the answer to Open Problem 1 is negative in both settings.*

*Proof.* All four Hilbert spaces in the statement are separable and infinite-dimensional. Lemma 1 gives  $\mathcal{M} \subseteq \mathcal{K}(X, Y)$  for every proper ideal  $\mathcal{M}$ .

For the Hardy space, the source chooses  $b \in H^\infty \setminus \text{VMO}$ ; then  $H_b = 0$  but  $H_{\bar{b}}$  is not compact. For the Bergman space on the disk, the source uses an explicit infinite Blaschke product with the same two properties, and it extends this example by  $b(z) = b(z_1)$  to  $\mathbb{B}_n$  [1, Example 2]. In every case,

$$H_b = 0 \in \mathcal{M}, \quad H_{\bar{b}} \notin \mathcal{K}(X, Y) \supseteq \mathcal{M}.$$

Therefore the defining equivalence for BCP fails. Since  $\mathcal{M}$  was arbitrary, no proper operator ideal works.  $\square$

## 5 Scope, novelty check, and review recommendation

The theorem answers exactly Open Problem 1 under the standard two-sided ideal meaning used for compact and Schatten ideals. If “ideal” were intended to mean a symbol-dependent class lacking stability under arbitrary left and right bounded multiplication, Lemma 1 would not apply; such a class is not an operator ideal in the usual sense. The full class  $\mathcal{B}(X, Y)$  has BCP trivially and is precisely the excluded trivial case.

On 11 August 2026, the run registry and all cheap indexes had no prior answer for arXiv:2304.09716 or this ideal-factorization route. Exact-title, exact- problem, author, Hardy/Bergman operator-ideal, and Berger–Coburn searches of the local arXiv corpus and the web found the source and later work on doubling Fock spaces and symmetrically normed ideals, but no answer to Open Problem 1. The later doubling-Fock paper arXiv:2312.06656 explicitly makes partial progress on a different open problem from the source. This is a bounded novelty check, not an exhaustive priority claim.

Human review should focus on the intended meaning of “operator ideal” in Open Problem 1. Once the standard two-sided meaning is confirmed, the factorization in Lemma 1 and the source’s noncompactness examples make the conclusion immediate.

## References

- [1] Z. Hu and J. A. Virtanen, *On the Berger–Coburn phenomenon*, Operator Theory: Advances and Applications 297 (2024), 223–234; arXiv:2304.09716.